PROPOSAL FOR ORGANIZING AN INTERNATIONAL CONFERENCE

"International Conference on Membrane Technology, Desalination, Artificial Intelligence and Environmental Sustainability (MemDesAI-2027)"

Jointly Organized by

Indian Institute of Technology Roorkee

&

CSIR – Indian Institute of Petroleum (CSIR-IIP), Dehradun

Proposed Dates: 6th to 8th December 2027 | Venue: IIT Roorkee, Uttarakhand, India

Role	Name & Affiliation
Convener	Dr. Bhaskar Jyoti Deka, Associate Professor, Department of Hydrology, IIT Roorkee (Joint Faculty, Centre for Nanotechnology)
Co-Convener	Dr. Racha Surya Murali, Principal Scientist, Adsorption and Membrane Separation, CSIR-IIP Dehradun Dr. Anshul Yadav, Assistant Professor, Water Resources Development and Management, IIT Roorkee
Host Institution	Indian Institute of Technology Roorkee
Partner Institution	CSIR – Indian Institute of Petroleum, Dehradun
In-collaboration	Department of Chemical and Biological Engineering, Hong Kong University of Science & Technology

1. Background and Rationale

Membrane-based separation has emerged as one of the most energy-efficient platforms for water purification and resource recovery, and it now sits at the intersection of three accelerating trends: the global push for affordable, low-carbon desalination; the adoption of artificial intelligence and machine learning to design, monitor, and optimize separation processes; and the growing demand for environmentally sustainable materials and processes across the membrane life cycle. Advances in nanocomposite membranes, membrane distillation, forward and reverse osmosis, and gas separation are increasingly coupled with data-driven approaches for fault detection, fouling prediction, and process control, while circular-economy thinking is reshaping how membranes are manufactured, used, and retired. Research on these fronts, however, remains fragmented: materials scientists, process engineers, data scientists, and environmental researchers rarely share a common platform to align priorities and pool resources.

IIT Roorkee's Department of Hydrology, together with its Centre for Nanotechnology and International Centre of Excellence for Dams, has built a strong research base in desalination, membrane distillation, nanocomposite membrane fabrication, and computational modelling of membrane processes, evidenced by sustained funding from ISRO, DST, ANRF, and international partners in Germany, Austria,

Hong Kong, and South Korea. CSIR-IIP Dehradun brings complementary strength in petroleum refining, catalysis, and industrial-scale membrane applications, along with experience in process modelling and plant-level data analytics. A joint international conference between the two institutions is therefore timely: it will consolidate India's position in membrane science, create a structured dialogue between academic researchers, national laboratories, and industry on the convergence of membrane technology, desalination, AI, and sustainability, and directly support the Government of India's priorities under the Jal Jeevan Mission, the National Hydrogen Mission, the National AI Mission, and Atmanirbhar Bharat in advanced materials manufacturing.

2. Conference Theme and Scope

The conference is framed around four interlocking pillars: the fundamental science and engineering of membrane materials; desalination and water treatment as the sector's largest-scale application; artificial intelligence as an emerging tool for membrane design, monitoring, and process optimization; and environmental sustainability as a cross-cutting imperative spanning circular-economy manufacturing, pollutant capture, and life-cycle impact. Each pillar will be represented as a dedicated technical track, with cross-cutting sessions designed to surface shared materials, characterization methods, and scale-up challenges.

Proposed Technical Tracks

- Membrane Technology: Nanocomposite and thin-film composite membranes, membrane distillation, forward and reverse osmosis, gas separation, biomembranes, and novel fabrication and characterization techniques.
- Desalination: Seawater and brackish-water desalination, brine management and resource recovery, hybrid and renewable-energy-coupled desalination systems, and recovery of critical minerals and rare earth elements.
- Artificial Intelligence: AI/ML-assisted membrane material design, fouling and fault prediction, digital twins and process control, and computational and CFD modelling of membrane processes.
- Environmental Sustainability: Membrane-based capture of emerging contaminants and micro/nanoplastics, air filtration and CO_2 separation, circular-economy approaches to membrane manufacturing and end-of-life, and life-cycle assessment of membrane processes.

Cutting across these tracks, dedicated sessions will address industry-academia pathways for technology transfer and scale-up, and opportunities for translating AI-driven insights into deployable membrane systems — areas in which the organizing team already has active sponsored projects and international collaborations.

3. Objectives

- Create an international forum bringing together academia, national laboratories, and industry working across membrane technology, desalination, artificial intelligence, and environmental sustainability.

- Showcase and benchmark Indian membrane research against global developments through keynote and invited lectures from leading international laboratories.
- Catalyze new collaborative research proposals, joint publications, and bilateral/multilateral funding partnerships, including with DST, ANRF, IGSTC, and equivalent international agencies.
- Provide early-career researchers, doctoral scholars, and postdoctoral fellows a platform to present their work, receive mentorship, and build professional networks.
- Strengthen industry linkages to translate laboratory-scale membrane and AI-driven process innovations into deployable, commercially viable, and environmentally sustainable technologies.

4. Organizing Structure

The conference will be steered by a Joint Organizing Committee comprising faculty from IIT Roorkee and scientists from CSIR-IIP Dehradun, supported by an International Advisory Committee drawn from the Convener's existing collaborator network — including the City University of Hong Kong, KAUST, and partner institutions in Germany, Austria, and South Korea — and a National Advisory Committee of senior Indian academicians and industry representatives.

Committee	Composition & Role
Patron Committee	Directors of IIT Roorkee and CSIR-IIP Dehradun, with senior DST/CSIR representation
Joint Organizing Committee	Convener and Co-Conveners, together with faculty and scientists from the Department of Hydrology, the Centre for Nanotechnology (IIT Roorkee), and the Membrane Separation & Catalysis groups (CSIR-IIP)
International Advisory Committee	1. Prof. Alicia Kyoung Jin An, Department of Chemical and Biological Engineering, Hong Kong University of Science & Technology; 2. Prof. Ho Kyong Shon, ARC Research Hub for Nutrients in a Circular Economy, Center for Technology in Water and Wastewater, School of Civil and Environmental Engineering, University of Technology Sydney, Australia 3. Dr. Volkan Filiz, Institute of Membrane Research, Helmholtz-Zentrum Hereon, Hamburg, Germany; 4. Prof. Dr.-Ing. Mark Hlawitschka, Johannes Kepler University Linz, Austria; 5. Prof. Sanghyun Jeong, Pusan National University, South Korea
National Advisory Committee	Senior faculty from IITs, NITs, CSIR and DRDO laboratories, along with representatives from ISRO, DST, and industry, including process and energy sector partners
Technical Program Committee	Track chairs for the Membrane Technology, Desalination, Artificial Intelligence, and Environmental Sustainability sessions, responsible for abstract review and session scheduling
Student Organizing	Doctoral and master's scholars from the Convener's research group and from

Team	CSIR-IIP, coordinating logistics, registration, and volunteer support

5. Conference Format and Tentative Schedule (3 Days)

The conference is proposed as a three-day event combining plenary and keynote sessions, parallel technical tracks, poster sessions, an industry exhibition, and a young researchers' forum, with an optional pre-conference tutorial/workshop day for hands-on training on membrane characterization techniques and AI/ML tools for process data analysis.

Day	Key Activities
Day 1	Inaugural session; plenary keynotes from international and national experts; track-wise parallel sessions on Membrane Technology (materials, fabrication, and characterization); poster session I; welcome reception
Day 2	Plenary and invited talks; parallel sessions on Desalination and Environmental Sustainability; industry panel discussion on technology transfer and scale-up; poster session II; conference banquet
Day 3	Parallel sessions on Artificial Intelligence for membrane design, process control, and diagnostics; young researchers' forum with best-paper and best-poster awards; panel discussion on future India–international collaboration; valedictory session

An estimated 250–300 delegates are anticipated, comprising roughly 30–40 international and national invited speakers, 150–180 contributed paper/poster presenters — including a significant share of doctoral and master's scholars — and representatives from funding agencies and industry.

6. Expected Outcomes and Deliverables

- A peer-reviewed conference proceedings volume, and a curated special issue in a Q1 journal (e.g., Desalination, Journal of Membrane Science, or equivalent) comprising extended versions of selected papers.
- Formal Memoranda of Understanding / Letters of Intent for joint research between IIT Roorkee, CSIR-IIP, and at least two to three international partner institutions.
- A white paper synthesizing research priorities and policy recommendations at the intersection of membrane technology, desalination, AI, and environmental sustainability, to be shared with DST, ANRF, and other relevant ministries.
- Strengthened industry engagement through at least one industry-sponsored session and exploratory discussions on pilot-scale technology transfer, including AI-enabled process monitoring.
- Enhanced visibility and international standing for the host departments and for Indian membrane science more broadly.

7. Budget Estimate

The budget below is indicative and structured so that support can be sought from multiple funding

sources — including DST, ANRF (SERB), CSIR, ISRO, the Indo-German Science & Technology Centre (IGSTC) or equivalent bilateral programmes, and industry sponsorship — in proportions to be finalized once specific calls are identified. Figures are in Indian Rupees (₹) and exclude any institutional in-kind contribution such as venue, manpower, and existing laboratory infrastructure.

Head	Estimated Cost (₹)	Remarks
Venue, audio-visual, and conference infrastructure	6,00,000	Auditorium, session halls, poster boards, live-streaming
Catering and hospitality (3 days, ~280 delegates)	12,00,000	Working lunches, refreshments, welcome reception, banquet
Conference proceedings, special-issue processing charges, and publications	5,00,000	Editing, ISBN/DOI, journal APCs for selected papers
Promotional material, website, conference kits, certificates	3,00,000	Website hosting, banners, bags, badges, kits
Student/volunteer support and local transport	2,50,000	Local conveyance, volunteer honoraria
Best paper/poster awards and young researcher forum	1,50,000	Cash awards/mementoes for 3 categories
Contingency (approx. 8%)	5,00,000	Unforeseen logistics and last-minute requirements
Total (Indicative)	**35,00,000**	To be apportioned across sponsoring agencies and institutional/industry contributions

Registration fees, differentiated for students, academic delegates, and industry participants, will be levied as a partial cost-recovery measure and adjusted against the final sanctioned grant.

8. Proposed Timeline

Milestone	Target Period
Finalize organizing committees, advisory boards, and funding applications	Q1 2027 (Jan–Mar)
Launch conference website; open call for abstracts and sponsorship	Q2 2027 (Apr–Jun)
Abstract review, speaker confirmation, and program finalization	Q3 2027 (Jul–Sep)
Registration closes; logistics, venue, and travel arrangements finalized	Oct 2027
Conference (3 days)	November 2027
Post-conference: proceedings compilation, special-issue submission, MoU follow-up	Dec 2027 – Q1 2028

9. Institutional Roles and Complementarity

IIT Roorkee will serve as the host institution, providing the venue and core organizing manpower drawn from the Department of Hydrology and Centre for Nanotechnology, and leading the Desalination and Environmental Sustainability tracks, building on its existing sponsored projects in desalination, nanocomposite membranes, and environmental hydrology. CSIR-IIP Dehradun will co-lead the Membrane Technology track, contributing its expertise in catalysis, refining, and industrial-scale membrane applications, and will support outreach to the petroleum and energy sector industry base. The Artificial Intelligence track will be jointly coordinated by both institutions, drawing on their existing computational modelling and process-data-analytics capabilities. This complementarity is intended to ensure the conference genuinely spans the fundamental-to-applied and academic-to-industrial spectrum implied by the membrane technology–desalination–AI–sustainability framing, rather than defaulting to a single institution's areas of strength.

10. Conclusion

This conference is proposed as a focused, outcome-oriented platform that consolidates the complementary strengths of IIT Roorkee and CSIR-IIP Dehradun in membrane science and technology, while opening a structured dialogue with leading international laboratories and domestic industry on the convergence of membrane technology, desalination, artificial intelligence, and environmental sustainability. Given the Convener's track record of successfully coordinating four prior international/bilateral workshops — MemDEW 2023, MemSci-Brine 2024, SHINE 2025, and MemTechAI 2026 — with support from ANRF and the Indo-German Science & Technology Centre, the organizing team is well positioned to build on the MemTechAI series and deliver this larger-scale event, translating it into lasting research collaborations, publications, and industry partnerships.